

**Russell J. Montjoy**  
**Phone: (720)234-5161**  
**Email: russellmontjoy@gmail.com**

---

## **QUALIFICATIONS:**

Highly knowledgeable spacecraft engineer with expertise supporting operations for United States Air Force and the private sector. Performed real-time on-orbit operations, spacecraft command/control, telemetry procedure development, command script development, training and operations procedure development. Possess excellent leadership and communication skills and the ability to foster an inclusive workplace where diversity and individual differences are value and leveraged to achieve the mission.

## **CORE SKILLS:**

Satellite Systems: GPS, M.S.X, UHF/FO, DSP, MILSTAR, FLTSAT, ACE  
Space Systems Operator Courses – U.S. Air Force  
(UHF/FO) (FLTSAT) Spacecraft Engineering – General Dynamics  
Programming languages: C++, python, JavaScript

## **PROFESSIONAL EXPERIENCE:**

Lockheed Martin Aerospace Company Aurora, CO. 2018-2023  
Satellite Systems Engineer - Vehicle Technical Advisor

- Understanding and experience with space segment to ground segment integration
- Develop and continually refine software tools to extract/interpret data and perform trending analysis on system hardware and software
- Analyzing, trending, and reporting the performance of subsystem during end-to-end testing, launch and orbit raising and post launch test activities.
- Perform real time troubleshooting and analysis during satellite anomalies, including real-time problem identification and resolution, support in case of any in-orbit anomaly and participate in the investigation of satellite anomalies
- Technical knowledge of spacecraft systems (bus, payload) and subsystems (attitude and control systems, propulsion, life support systems, etc.), ground communications and mission data processing systems, space vehicle mission planning and command and control systems
- Knowledge of orbital mechanics and associated analytical tools
- Experience with mission readiness process, control gates, and associated entry/exit criteria

Lockheed Martin Aerospace Company Springfield, V.A. 2015-2018  
*Satellite Subsystems Engineer –Power subsystems Engineer*

- Analyze technical data, verify and sustain specific system configurations in accordance with defined constraints and operational requirements.
- Demonstrated experience in spacecraft electrical power system (EPS) technology maturation, design, development, production, test, or sustainment.
- Perform calibration activities to ensure specification requirements are met.
- Create/maintain documentation and procedures associated with the system, interface with factory team and other support contractors.
- Provide technical briefings to the customer and company management
- Understanding of satellite design principles and various related disciplines such as the electrical power subsystem

Lockheed Martin Aeronautics Company Patuxent River Naval Air Station, M.D. 2011-2015  
*Electronics Engineer – Flight Test Data Operations (Program: Joint Strike Fighter-JSF)*

- Responsible for post-flight processing of aircraft test data, including recovery, computation, and display of eight aircrafts and two aircraft simulators. Includes aircraft, real time, IADS, TSPI (GPS), scribe and pirate data (weather).
- Interface with technical personnel in the diagnosis and resolution of problems encountered in the recovery of test data utilizing various data systems, including Windows and UNIX servers and specialized data processing equipment.
- Manage Interactive Analysis and Display System (IADS) configuration files for the Real-Time Telemetry Processing System (RTPS) pre and posttest, management of Test Data Requests (TDR)
- Executed the understanding of software development, verification, and validation.

RAYTHEON COMPANY, Aurora, Colorado  
GPS OCX Sr. Systems Engineer

2010 – 2011

- Developed and executed integrated organizational plans, policies and procedures. Guided the development of business and technical strategies, goals and objectives.
- Provided oversight and approval of technical approaches, products and processes.
- Established standards for CPT and directed sub-team leads and team members towards meeting goals.
- Maintained relationships and partnerships with customers, stakeholders, peers, partners and direct reports.
- Performed DOORs exports and converted extracted data to program deliverable documents (Item Detailed Specifications and Item Performance Specifications). Guiding documents through Engineering Review Board (ERB) process for base lining. These documents were vital for the successful completion CDR.

GENERAL DYNAMICS, Colorado Springs, Colorado  
Spacecraft Engineer

2007 – 2010

- Delivered key support to satellite mission planning activities and creation of spacecraft products.
- Supported on-console engineering activities, including verification of pass plans and on-orbit telemetry tracking, and maintained satellites within their orbit tolerances.
- Executes necessary ground limit changes to reflect current operating state of the spacecraft and long-term telemetry trends
- Providing technical advice and guidance to operations and evolving technologies and capabilities under development
- Evaluated health and status of FLTSAT and UHF/FO system vehicles. Prepared and delivered antenna viewing angle reports, solar and lunar eclipse reports, and sensor intrusion predictions.
- Initiated immediate responses to spacecraft System anomalies.

LINQUEST CORPORATION, Colorado Springs, Colorado  
GPS IIR/IIR-M Engineer

2005 – 2007

- Processed and analyzed raw measurement and anomaly data for the GPS satellite system. Developed results reports for both internal and external users.
- Ensured performance of GPS receiver systems by overseeing testing of vehicle-specific modifications, ground station changes, and crew training.
- Conducted spacecraft state-of-health monitoring and telemetry trending operations.
- Minimized resource conflicts by coordinating programmatic GPS activities with various agencies.
- Tested spacecraft power system hardware (solar arrays, batteries, power electronics, and other space Power subsystem elements).
- Develops trade studies, analyzes results, and performs design reviews

UNITED STATES AIR FORCE, Schriever AFB, Colorado  
Satellite System Operator/ Ground Systems Operator/ Orbital Analyst

1999 – 2005

- Planned normal commanding supports to maintain proper satellite performance, plan launch and early orbit activities for SBIRS assets.
- Conducted daily evaluation of the sensor payload and the spacecraft state of health.
- Executed TT&C and spacecraft ranging and commanding, during 24/7 rotating shift operations according to

Scheduled and dynamic operational requirements.

- Performs In/out-of-contact spacecraft telemetry trending and analysis to ensure continued health and safety of the spacecraft out of view of a ground station.
- Supported satellite operations center's (SOC) ground equipment/on-orbit satellites valued at \$5.6 billion.
- Identified a remote ground facility misconfiguration during a critical GPS contact-prevented loss of navigation data to war fighters.
- Planned and performed launch/early-orbit (LEO) operations from launch pad checkout through activation.
- Tested spacecraft power system hardware (solar arrays, batteries, power electronics, and other space Power subsystem elements)
- Identify, assess, and mitigate hazards present in spacecraft using the process described MIL-STD-882.
  - Plan, develop, and aid in the execution of new hazardous manufacturing and test procedures.
  - Generate design and test safety requirements for spacecraft systems and subsystems.

## **Education & Certifications**

Bachelor of Science in Computer Science – Colorado Technical University  
(2006)

Microsoft Azure Administrator (AZ-104) – April 2025

Microsoft Azure Fundamentals (AZ-900) – March 2025

Full Stack Development Certification – University of Denver Feb 2024